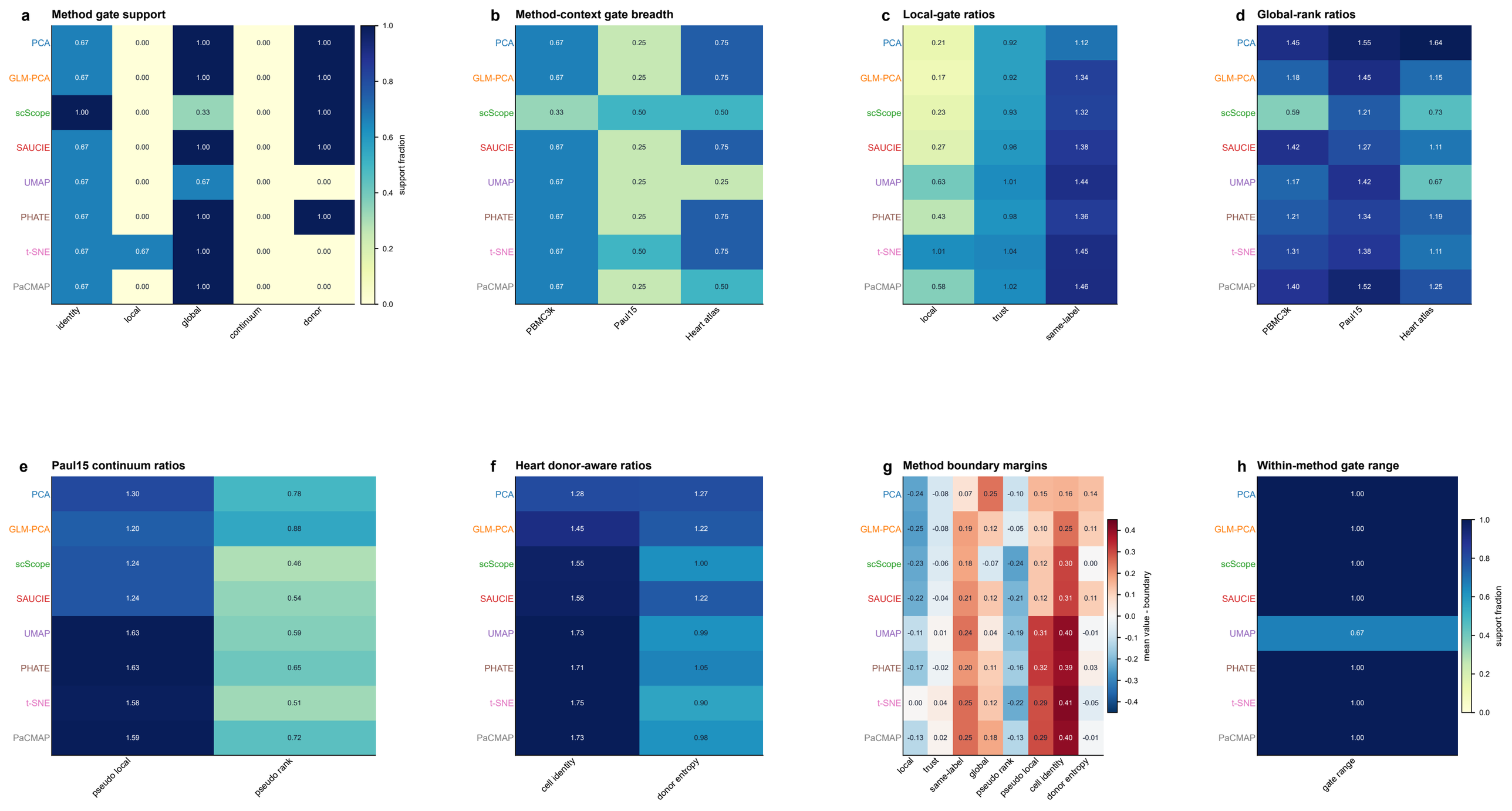
a
Method gate support
PCA 0.67 0.00 1.00 0.00 1.00
GLM-PCA 0.67 0.00 1.00 0.00 1.00
scScope 1.00 0.00 0.33 0.00 1.00
SAUCIE 0.67 0.00 1.00 0.00 1.00
UMAP 0.67 0.00 0.67 0.00 0.00
PHATE 0.67 0.00 1.00 0.00 1.00
t-SNE 0.67 0.67 1.00 0.00 0.00
PaCMAP 0.67 0.00 1.00 0.00 0.00
identity
local
global
continuum
donor
support fraction
1.0
0.8
0.6
0.4
0.2
0.0
b
Method-context gate breadth
PCA 0.67 0.25 0.75
GLM-PCA 0.67 0.25 0.75
scScope 0.33 0.50 0.50
SAUCIE 0.67 0.25 0.75
UMAP 0.67 0.25 0.25
PHATE 0.67 0.25 0.75
t-SNE 0.67 0.50 0.75
PaCMAP 0.67 0.25 0.50
PBMC3k
Paul15
Heart atlas
c
Local-gate ratios
PCA 0.21 0.92 1.12
GLM-PCA 0.17 0.92 1.34
scScope 0.23 0.93 1.32
SAUCIE 0.27 0.96 1.38
UMAP 0.63 1.01 1.44
PHATE 0.43 0.98 1.36
t-SNE 1.01 1.04 1.45
PaCMAP 0.58 1.02 1.46
local
trust
same-label
d
Global-rank ratios
PCA 1.45 1.55 1.64
GLM-PCA 1.18 1.45 1.15
scScope 0.59 1.21 0.73
SAUCIE 1.42 1.27 1.11
UMAP 1.17 1.42 0.67
PHATE 1.21 1.34 1.19
t-SNE 1.31 1.38 1.11
PaCMAP 1.40 1.52 1.25
PBMC3k
Paul15
Heart atlas
e
Paul15 continuum ratios
PCA 1.30 0.78
GLM-PCA 1.20 0.88
scScope 1.24 0.46
SAUCIE 1.24 0.54
UMAP 1.63 0.59
PHATE 1.63 0.65
t-SNE 1.58 0.51
PaCMAP 1.59 0.72
pseudo local
pseudo rank
f
Heart donor-aware ratios
PCA 1.28 1.27
GLM-PCA 1.45 1.22
scScope 1.55 1.00
SAUCIE 1.56 1.22
UMAP 1.73 0.99
PHATE 1.71 1.05
t-SNE 1.75 0.90
PaCMAP 1.73 0.98
cell identity
donor entropy
g
Method boundary margins
PCA -0.24 -0.08 0.07 0.25 -0.10 0.15 0.16 0.14
GLM-PCA -0.25 -0.08 0.19 0.12 -0.05 0.10 0.25 0.11
scScope -0.23 -0.06 0.18 -0.07 -0.24 0.12 0.30 0.00
SAUCIE -0.22 -0.04 0.21 0.12 -0.21 0.12 0.31 0.11
UMAP -0.11 0.01 0.24 0.04 -0.19 0.31 0.40 -0.01
PHATE -0.17 -0.02 0.20 0.11 -0.16 0.32 0.39 0.03
t-SNE 0.00 0.04 0.25 0.12 -0.22 0.29 0.41 -0.05
PaCMAP -0.13 0.02 0.25 0.18 -0.13 0.29 0.40 -0.01
local
trust
same-label
global
pseudo rank
pseudo local
cell identity
donor entropy
mean value - boundary
0.4
0.3
0.2
0.1
0.0
-0.1
-0.2
-0.3
-0.4
h
Within-method gate range
PCA 1.00
GLM-PCA 1.00
scScope 1.00
SAUCIE 1.00
UMAP 0.67
PHATE 1.00
t-SNE 1.00
PaCMAP 1.00
gate range
support fraction
1.0
0.8
0.6
0.4
0.2
0.0